

ECA5612 –Advanced Wireless Communication Systems 5G / 6G Communication Systems

Experiment: 7-Analog Beamforming Analysis Using MATLAB

Objective – 1 Analyze Beamforming Gain (dB)

Program:

```
clc;
```

```
clear;
```

```
close all;
```

```
antennas = [2 4 8 16 32]; % Number of Antenna Elements
```

```
gain = 10*log10(antennas); % Beamforming Gain (dB)
```

```
figure
```

```
bar(antennas, gain)
```

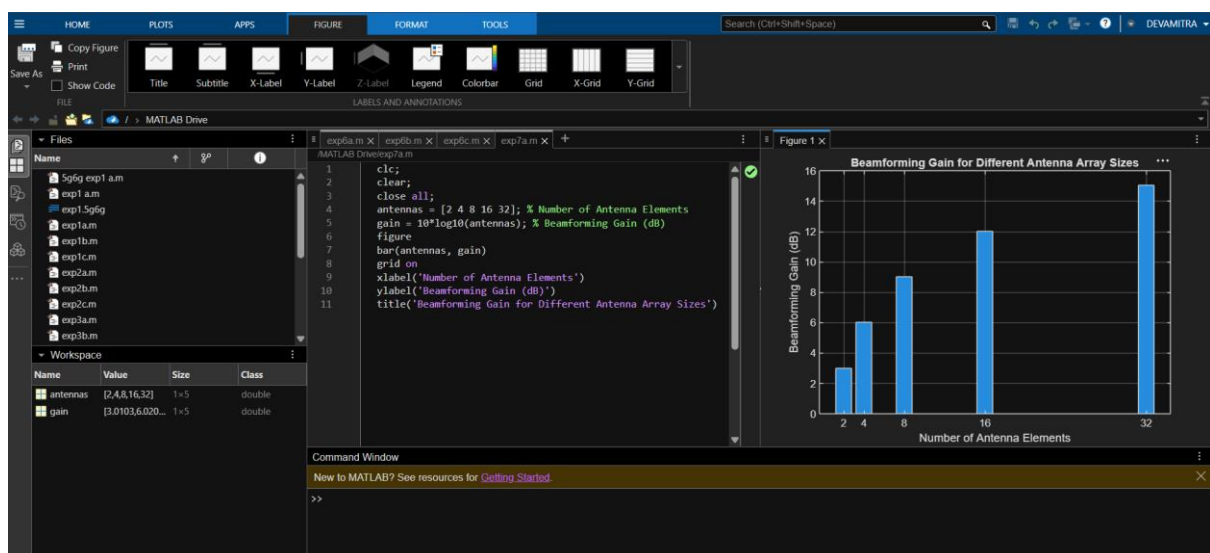
```
grid on
```

```
xlabel('Number of Antenna Elements')
```

```
ylabel('Beamforming Gain (dB)')
```

```
title('Beamforming Gain for Different Antenna Array Sizes')
```

Output:



Objective – 2 Compare Beam Width (Degrees)

Program:

```
clc;

clear;

close all;

antennas = [2 4 8 16 32]; % Number of Antenna Elements

beamwidth = [90 45 22.5 11.25 5.625]; % Beam Width (degrees)

figure

plot(antennas, beamwidth,'o-','LineWidth',2)

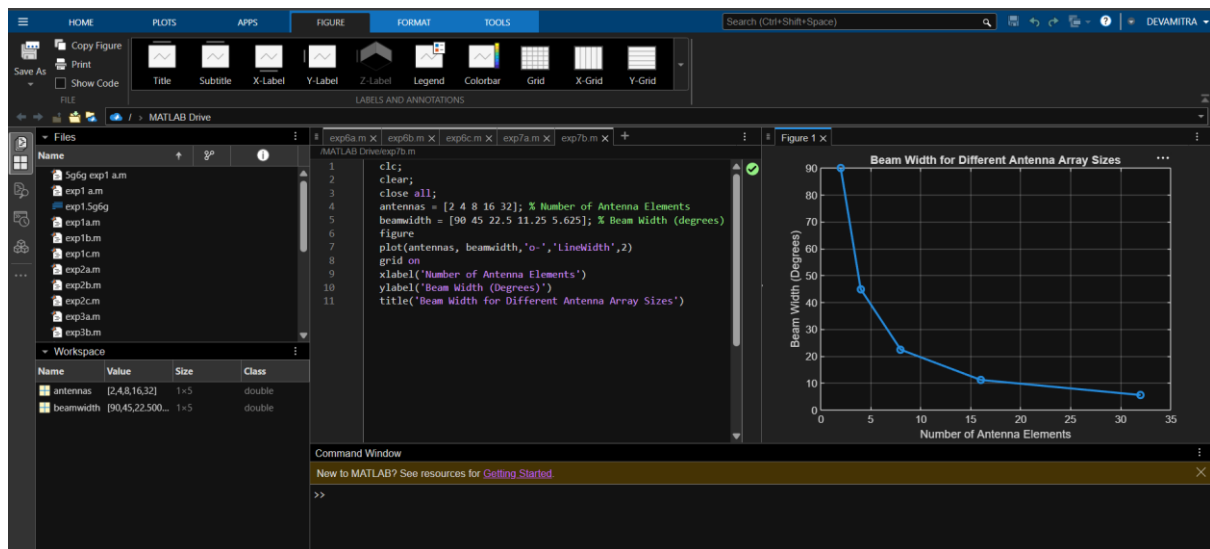
grid on

xlabel('Number of Antenna Elements')

ylabel('Beam Width (Degrees)')

title('Beam Width for Different Antenna Array Sizes')
```

Output:



Objective – 3 Analyze Main Lobe Direction (Degrees)

Program:

```
clc;

clear;

close all;

theta = [-60 -30 0 30 60]; % Main Lobe Direction (degrees)

gain = [1 1 1 1 1]; % Constant normalized gain

figure

polarplot(deg2rad(theta), gain, 'o', 'LineWidth', 2)

title('Main Lobe Direction in Analog Beamforming')
```

Output:

